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VII SUMMER INTERNSHIP (CS376SI):Review-1

Lightweight AI-Driven Multi-Objective Optimization for Renewable and Carbon Aware Cloud Scheduling

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About

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Organization

Women in Cloud (WiC) Center of Excellence India is dedicated to empowering students through advanced cloud technologies, fostering innovation, entrepreneurship, and strong industry collaboration.

Domain

Cloud-Native Application Development and Modern Cloud Solutions

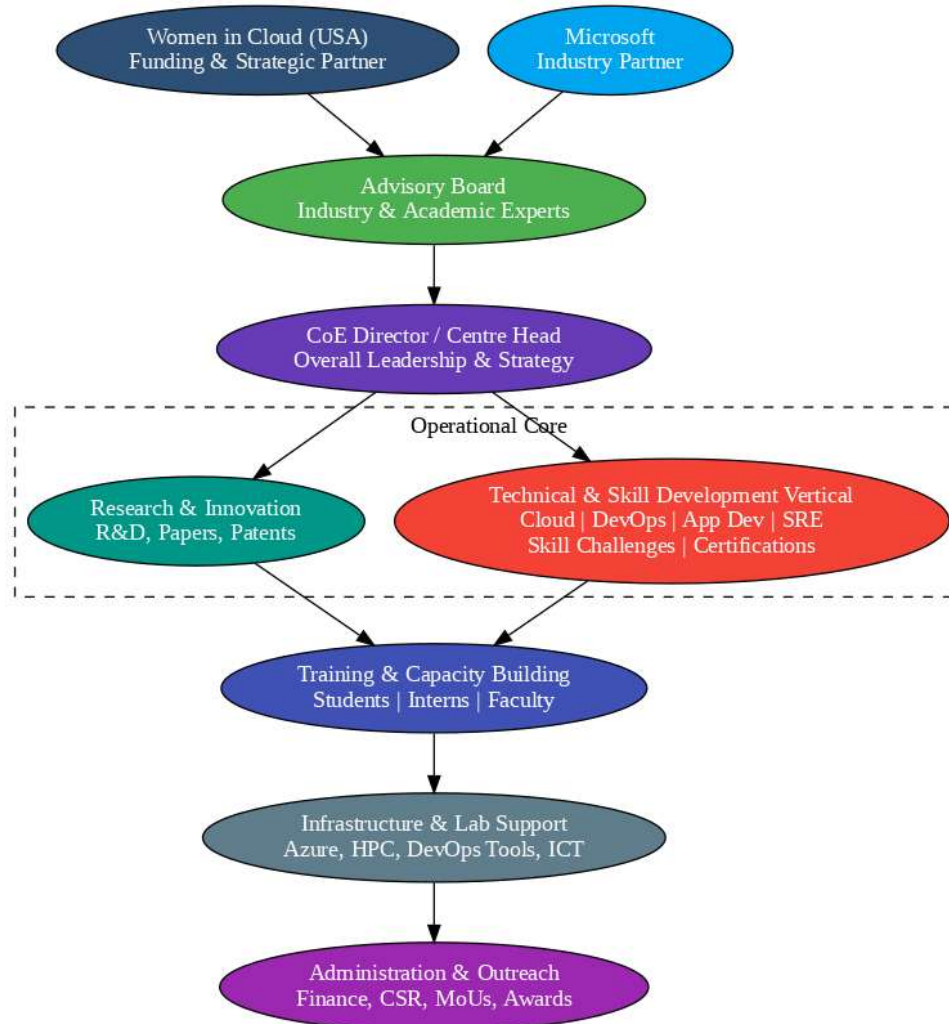




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Understanding Organisation Structure

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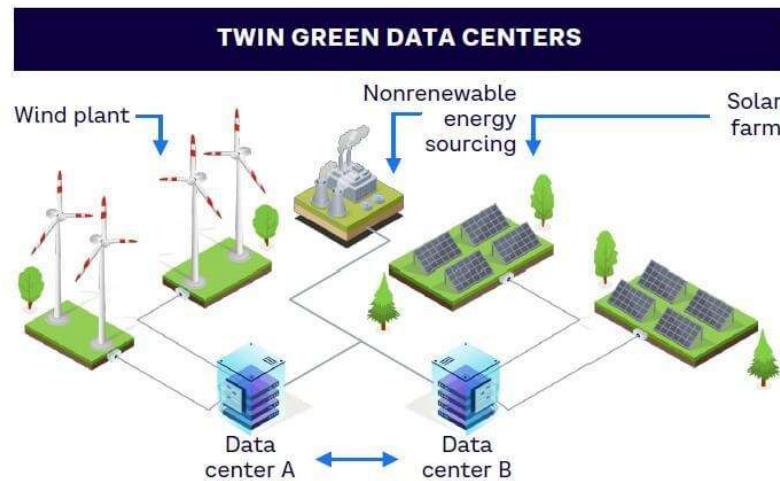




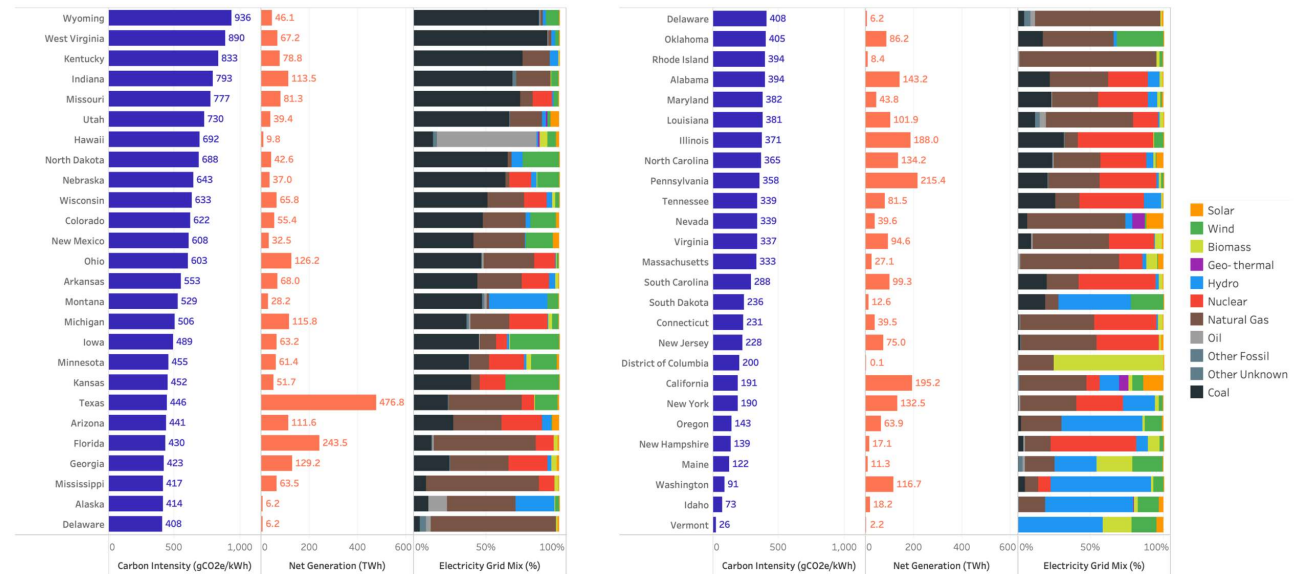
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Understanding Industry Operations



United States - Electricity Grid Carbon Intensity (gCO₂e / kWh)



PRIOPTA

(Data Source: [EPA eGrid 2018](#). Visualization by Priopta)



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Application of Engineering Knowledge

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☁️ Cloud Computing

- Optimized workload placement across geographically distributed cloud regions
- Maintained performance, scalability, and load balance



Machine Learning

- Random Forest regression for carbon intensity & renewable energy forecasting
- Used historical grid and meteorological datasets



Algorithms

- Modeled scheduling as a multi-objective optimization problem
- Applied NSGA-II to minimize carbon emissions, maximize renewable usage, and balance workloads



Data Engineering

- Built automated pipelines to fetch data from carbon intensity & weather APIs
- Performed preprocessing, normalization, and continuous data flow into the scheduling engine



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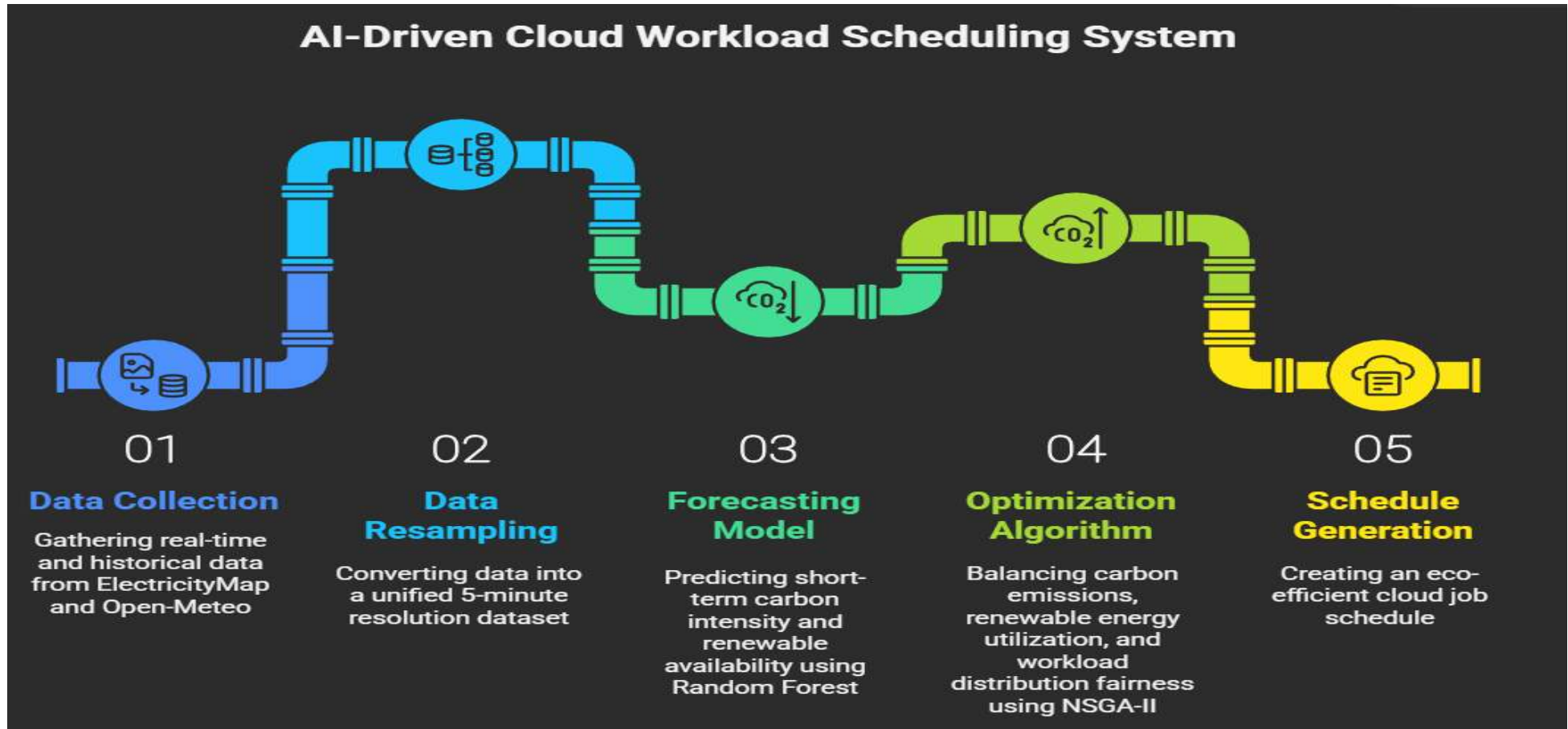
Software Requirements

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- Programming Language: Python 3.x
 - Libraries: Pandas, NumPy, Scikit-learn, DEAP, Request, PyMOO, and Plotly
 - Visualization Framework: Streamlit
 - APIs: ElectricityMap (carbon intensity), Open-Meteo (weather data)
 - Development Tools: VS Code, Google Colab Notebook



Methodology & System Workflow





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